

The Fabius Function and Its Inverse are Not Elementary

Density of the analytic locus under algebraic branches and inversion

Human-readable account of the formal development

Vladimir Reshetnikov's ProveIt repository

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Abstract

The Fabius function F is a C^∞ probability distribution function on $[0, 1]$ that is real analytic at no point of that interval. We prove, and formalize in Lean 4 on top of `MathLib`, that no elementary function of one real variable agrees with F on any subset of $[0, 1]$ with nonempty interior; in particular the restriction of F to $(0, 1)$ is not elementary.

Since F is C^∞ , no smoothness obstruction can decide the question: the obstruction must be analyticity. The whole content of the argument is therefore a single positive statement about the class of elementary functions, which we prove in the form we need it and no stronger:

the set of points at which an elementary function is real analytic is a dense open subset of $\mathbb{R}$.

Density cannot in general be replaced by analyticity everywhere: $x \mapsto x^{-1}$ and $x \mapsto |x|$ are elementary and fail to be analytic at the origin. The induction that proves it is short but has one genuinely delicate step, the unary one, where a locally constant intermediate value must be separated from a locally nonconstant one; the separation is furnished by the isolated-zeros theorem for real analytic functions of one variable.

The same conclusion is then proved for two classes the inductive definition does not reach. The first consists of continuous branches, on a nonempty open set, of polynomial equations whose coefficients are elementary and whose leading coefficient is nonzero throughout that set. This covers continuous algebraic branches, including examples not expressible by radicals, and rests on a degree induction that avoids the discriminant, of which `MathLib` has no usable theory, in favour of the analytic implicit function theorem, of which it does.

The second concerns inverse branches. We prove that F^{-1} , which exists because F is a homeomorphism of $[0, 1]$, is itself analytic at no point of $[0, 1]$ and hence not elementary either; and then that adjoining continuous inverse branches which are analytic off their open branch domains, at any depth, preserves density of the analytic locus. The organizing observation is that the one-sided identity $h(g(x)) = x$ makes g a right inverse of h and an implicit branch of the simplest possible equation, $h(z) - x = 0$, so the analytic inverse theorem is the implicit function theorem already in hand.

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1 Statement

1.1 The two inputs

Write $\mathcal{A}(f) = \{x \in \mathbb{R} : f \text{ is real analytic at } x\}$ for the *analytic locus* of $f: \mathbb{R} \rightarrow \mathbb{R}$. The proof combines one negative fact about the Fabius function with one positive fact about elementary functions.

Theorem 1.1 (Nowhere analyticity; already formalized). *Let F be the Fabius function, extended by 0 to the left of 0 and by 1 to the right of 1. Then $\mathcal{A}(F) = \mathbb{R} \setminus [0, 1]$. In particular F is real analytic at no point of $[0, 1]$, endpoints included.*

This is the unnumbered corollary to equation (24) of Arias de Reyna, *Arithmetic of the Fabius function* (arXiv:1702.06487v3), transferred from Rvachev’s up to F and completed at the right endpoint; in the repository it is `Fabius.canonical_fabius_analyticAt_iff` in the module `FabiusFunction.NowhereAnalytic`. Nothing in the present document reproves it.

Theorem 1.2 (Density of the analytic locus). *If $f: \mathbb{R} \rightarrow \mathbb{R}$ is elementary, then $\mathcal{A}(f)$ is a dense open subset of $\mathbb{R}$.*

Corollary 1.3 (Main result). *Let $U \subseteq [0, 1]$ have nonempty interior and let $g: \mathbb{R} \rightarrow \mathbb{R}$ be elementary. Then g and F do not agree on U . In particular there is no elementary function that agrees with F on $(0, 1)$, and F itself is not elementary.*

Proof. By [Theorem 1.2](#) the set $\mathcal{A}(g)$ is dense, so it meets the nonempty open set $\text{int } U$; pick $x \in \text{int } U \cap \mathcal{A}(g)$. If g and F agreed on U they would have the same germ at x , because $\text{int } U$ is a neighbourhood of x contained in U ; analyticity at a point depends only on the germ, so F would be analytic at $x \in [0, 1]$, contradicting [Theorem 1.1](#). $\square$

Remark 1.4. Hypothesising nonempty *interior* rather than openness costs nothing in the proof and buys a little: U may be a closed interval, or an interval with a nowhere dense closed set removed, or any set that is merely fat somewhere. (Removing *all* the rationals would not do: the irrationals in

an interval have empty interior, and the hypothesis genuinely fails.) What cannot be weakened is fatness itself: on a finite set an elementary function certainly can agree with F , since a polynomial interpolates any finite table of values.

Corollary 1.3 is the strengthened form used here of the assertion “the Fabius function is not elementary”, and it is strictly stronger than the bare non-elementarity of F as a function on $\mathbb{R}$. The bare statement leaves open the possibility that some elementary function agrees with F on all of $(0, 1)$ and differs from it outside; Corollary 1.3 rules that out, and rules it out on every subinterval.

Remark 1.5. Only a very weak consequence of Theorem 1.2 is actually used: that an elementary function is analytic at *at least one* point of every nonempty open set. Density is nevertheless the statement one must prove, because it is the statement that survives the induction: “analytic somewhere” is not preserved by intersection, whereas “dense open” is.

2 The class of elementary functions

2.1 Definition

We follow the description at https://en.wikipedia.org/wiki/Elementary_function: elementary functions are those obtained from polynomials, roots, exponentials, logarithms, and the trigonometric functions and their inverses by finitely many applications of the field operations and composition. Formally:

Definition 2.1 (Elementary functions). The class $\mathcal{E}$ of *elementary functions* $\mathbb{R} \rightarrow \mathbb{R}$ is the smallest class of functions such that

1. every constant function $x \mapsto c$ lies in $\mathcal{E}$, and the identity $x \mapsto x$ lies in $\mathcal{E}$;
2. if $f, g \in \mathcal{E}$ then $x \mapsto f(x) + g(x)$, $x \mapsto f(x)g(x)$ and $x \mapsto -f(x)$ lie in $\mathcal{E}$;
3. if $f \in \mathcal{E}$ then $x \mapsto f(x)^{-1}$ lies in $\mathcal{E}$;
4. if $f \in \mathcal{E}$ and $r \in \mathbb{R}$ then $x \mapsto f(x)^r$ lies in $\mathcal{E}$;
5. if $f \in \mathcal{E}$ then each of $\exp \circ f$, $\log \circ f$, $\sin \circ f$, $\cos \circ f$, $\arcsin \circ f$, $\arctan \circ f$ lies in $\mathcal{E}$.

Two features of Definition 2.1 deserve comment.

Composition is a theorem, not an axiom. Definition 2.1 contains no clause “ $f, g \in \mathcal{E} \Rightarrow f \circ g \in \mathcal{E}$ ”. It does not need one.

Proposition 2.2. $\mathcal{E}$ is closed under composition.

Proof. Induct on the derivation of the outer function f . Every clause of Definition 2.1 commutes with precomposition by a fixed g : the constant $x \mapsto c$ composed with g is again that constant; the identity composed with g is g itself, which is elementary by hypothesis; and $(f_1 + f_2) \circ g = (f_1 \circ g) + (f_2 \circ g)$, $(\exp \circ f) \circ g = \exp \circ (f \circ g)$, and likewise for every remaining clause. $\square$

Consequently $\mathcal{E}$ contains every function one expects: differences $f - g = f + (-g)$, quotients $f/g = f \cdot g^{-1}$, natural powers, all polynomial and rational functions, $\sqrt{\cdot}$ and the n -th roots, the absolute value $|x| = \sqrt{x^2}$, the tangent $\sin / \cos$, the hyperbolic functions $\sinh x = (e^x - e^{-x})/2$, $\cosh$, $\tanh$, the arccosine $\pi/2 - \arcsin$, and the inverse hyperbolic sine $\operatorname{arsinh} x = \log(x + \sqrt{1 + x^2})$.

Total functions and junk values. Every member of $\mathcal{E}$ is a *total* function $\mathbb{R} \rightarrow \mathbb{R}$. This is a deliberate choice, and it is the choice that makes Corollary 1.3 strongest. The formalization inherits Mathlib’s conventions

$$0^{-1} = 0, \quad \log x = \log |x|, \quad \log 0 = 0, \quad 0^r = 0 \ (r \neq 0), \quad 0^0 = 1,$$

$$x^r = |x|^r \cos(\pi r) \ (x < 0), \quad \arcsin x = \pm \frac{\pi}{2} \ (\pm x \geq 1),$$

so that clauses (3)–(5) of Definition 2.1 never fail to produce a function.

One might worry that junk values weaken the theorem, by admitting “elementary” functions that are not classically elementary. They do the opposite. Admitting more functions into $\mathcal{E}$ makes the assertion “ $g \in \mathcal{E}$ and $g = F$ on U ” easier to satisfy, hence its refutation stronger. And nothing is lost in the other direction: if a classical elementary expression is well formed at every point of an open set U — no vanishing denominator, no nonpositive logarithm, no real power of a nonpositive base, no arcsine of a value outside $[-1, 1]$ — then on U it agrees pointwise with the total function produced by the same derivation, since on U the junk values are never consulted. So every such classical elementary function on U is the restriction to U of a member of $\mathcal{E}$, and Corollary 1.3 applies to it.

Warning 2.3 (Odd roots need a different derivation). The condition “no real power of a nonpositive base” in the box above cannot be dropped, and it is the one place where the total-function convention bites. Mathlib’s x^r at a negative base is $|x|^r \cos(\pi r)$, so $(-8)^{1/3} = 2 \cos(\pi/3) = 1$, whereas the classical real cube root of -8 is -2 . The clause $x \mapsto f(x)^r$ of Definition 2.1 therefore supplies the n -th roots only on $[0, \infty)$.

Nothing is lost, because the classical real odd root is in $\mathcal{E}$ by a *different* derivation,

$$x \mapsto \frac{x}{|x|} \cdot |x|^{1/n},$$

which uses only division, the absolute value and a real power, and which evaluates to 0 at $x = 0$ under the same conventions. This is `IsElementary.signedRpow`, and `Fabius.signedRoot_pow` verifies formally that for odd n its n -th power is the identity, so it really is the real n -th root and not merely a plausible substitute.

2.2 What is not claimed

$\mathcal{E}$ contains Mathlib’s total real-power representatives of the principal n -th roots on $[0, \infty)$ for positive n and, by Warning 2.3, the classical odd-root functions on all of $\mathbb{R}$. It is not, as defined, closed under passage to an arbitrary algebraic function: Definition 2.1 has no clause producing a continuous branch of $P(x, y) = 0$ for a polynomial P with elementary coefficients whose Galois group is not solvable, and Liouville’s differential-field definition of “elementary” does admit such branches. Section 5 closes that gap, not by enlarging $\mathcal{E}$ but by proving the conclusion of Corollary 1.3 directly for such branches. What remains genuinely outside the result is stated there.

Two smaller gaps are worth naming here.

The first is that $\mathcal{E}$ is a class of *total* functions, so a classical elementary expression enters it only through a derivation whose junk values are never consulted on the region of interest; Warning 2.3 records the one place where the obvious derivation is the wrong one.

The second concerns variable exponents. Definition 2.1 provides $x \mapsto f(x)^r$ for a *fixed* real r , and `IsElementary.rpow_of_pos` supplies f^g whenever the base f is positive *at every real* — which covers $(1 + x^2)^{\sin x}$ and any base that is everywhere positive (no uniform lower bound is needed), but not $x \mapsto x^x$, whose base is positive only on $(0, \infty)$. That case is handled instead by the region statement below.

The sign of an everywhere nonvanishing base is not an obstruction. `Mathlib`'s x^y is sign-dependent at a negative base — there $x^y = |x|^y \cos(\pi y)$ — but a single elementary formula covers both signs at once,

$$f^g = \exp(\log f \cdot g) \cdot \cos\left(g\pi \cdot \frac{1-f/|f|}{2}\right),$$

since $(1 - f/|f|)/2$ is the indicator of $f < 0$, making the cosine $\cos 0 = 1$ where $f > 0$ and $\cos(g\pi)$ where $f < 0$; no absolute value is needed inside the logarithm because `Mathlib` already sets $\log x = \log |x|$. This is `IsElementary.rpow_of_ne_zero`, and it needs only that the base never vanishes. This is a sufficient uniform closure rule, not a claim that every power whose base has a zero is non-elementary: fixed natural powers are already covered by other constructors. At $f(x) = 0$ the conventions $0^r = 0$ for $r \neq 0$ and $0^0 = 1$ are discontinuous in the exponent, so the nonvanishing formula does not extend uniformly there. Nothing is lost for Corollary 1.3, and this is formalized rather than asserted: `Fabius.eqOn_rpow_of_pos` states that on any set where the base is positive, f^g agrees pointwise with $\exp(\log f \cdot g)$, which is a member of $\mathcal{E}$ whenever f and g are. Since Corollary 1.3 already rules out agreement on *every* nonempty open subset of $[0, 1]$, and the base of a classical variable-exponent expression is positive wherever the expression is well formed, such an expression is excluded too.

We also make no claim about elementary functions of a complex variable, and none about the closely related but distinct question of whether F satisfies an algebraic differential equation.

3 Analyticity of the generators

Each generator of Definition 2.1 is real analytic away from a finite set of arguments. All of these facts are in `Mathlib`, most of them in the form “ C^n for every n in $\mathbb{N}_0 \cup \{\infty, \omega\}$ ”; specializing to the analytic exponent ω and applying `ContDiffAt.analyticAt` converts them to the statements below.

Lemma 3.1.

1. $\exp, \sin, \cos$ and $\arctan$ are real analytic at every point of $\mathbb{R}$.
2. $\log$ is real analytic at every $x \neq 0$ — at negative arguments as well, because $\log$ is even under the convention $\log x = \log |x|$.
3. For each fixed $r \in \mathbb{R}$, the function $x \mapsto x^r$ is real analytic at every $x \neq 0$; at negative x this uses the identity $x^r = |x|^r \cos(\pi r)$.
4. $\arcsin$ is real analytic at every $x \notin [-1, 1]$; outside $[-1, 1]$ it is locally constant.
5. $x \mapsto x^{-1}$ is real analytic at every $x \neq 0$.

Remark 3.2. The bad sets are sets of *values* of the inner function, not sets of points of the domain — $\{0\}$ for the reciprocal, the logarithm and a real power, $\{-1, 1\}$ for the arcsine, and $\emptyset$ for the rest. This is the shape the induction consumes, and it is why Lemma 4.2 below is stated with a finite set $B \subseteq \mathbb{R}$ of forbidden values.

4 Density of the analytic locus

4.1 Openness

Lemma 4.1. For every $f: \mathbb{R} \rightarrow \mathbb{R}$ the set $\mathcal{A}(f)$ is open.

Proof. Analyticity at x means that f agrees near x with the sum of a convergent power series; that series then converges on a whole ball around x and, after a change of origin, represents f near every point of that ball. So $\mathcal{A}(f)$ contains a neighbourhood of each of its points. $\square$

4.2 The unary step

The following lemma is where the argument is actually made. Recall the shape of the difficulty. If f is analytic on a dense open set and g is analytic everywhere except at finitely many values, one would like to say that $g \circ f$ is analytic wherever f is analytic and $f(x)$ avoids the bad values. That set need not be dense: consider $f \equiv 0$ and $g(y) = y^{-1}$. What saves the statement is that in that degenerate case $g \circ f$ is constant, hence analytic anyway. Separating the two cases is exactly the isolated-zeros theorem.

Lemma 4.2 (Key lemma). *Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ and let $B \subseteq \mathbb{R}$ be finite. Suppose $\mathcal{A}(f)$ is dense and g is analytic at every $y \notin B$. Then $\mathcal{A}(g \circ f)$ is dense.*

Proof. Let U be a nonempty open set; we must produce a point of $U \cap \mathcal{A}(g \circ f)$. Since $\mathcal{A}(f)$ is dense there is $x_0 \in U \cap \mathcal{A}(f)$, and by Lemma 4.1 the set $V = U \cap \mathcal{A}(f)$ is an open neighbourhood of x_0 . Distinguish two cases.

Case 1: f is constant equal to some $c \in B$ on a neighbourhood of x_0 . Then $g \circ f$ equals the constant $g(c)$ on that neighbourhood, so $g \circ f$ is analytic at x_0 itself, and $x_0 \in U$.

Case 2: for no $c \in B$ is f eventually equal to c near x_0 . Fix $c \in B$. The function $f - c$ is analytic at x_0 , so by the isolated-zeros theorem for analytic functions of one real variable, either $f - c$ vanishes identically near x_0 — excluded in this case — or $f - c$ is nonvanishing on some punctured neighbourhood of x_0 . Thus for each of the finitely many $c \in B$ there is a punctured neighbourhood of x_0 on which $f \neq c$. Intersecting these finitely many punctured neighbourhoods with the punctured neighbourhood $V \setminus \{x_0\}$ leaves a punctured neighbourhood of x_0 , which is nonempty because $\mathbb{R}$ has no isolated points. Any point t of it satisfies $t \in U$, f analytic at t , and $f(t) \notin B$; hence g is analytic at $f(t)$ and $g \circ f$ is analytic at t . $\square$

Remark 4.3. The case distinction is not an artefact of the write-up. Without it the lemma is false as stated, for the naive reason above. What makes the two cases exhaustive is the isolated-zeros theorem, applied to $f - c$: an analytic germ either vanishes identically or vanishes nowhere on some punctured neighbourhood. A merely C^∞ inner function admits neither alternative — it can equal c on a sequence accumulating at x_0 without being constant there — so analyticity of f , and not just smoothness, is what the step consumes. Within the induction of this section it is the only place where a theorem of substance about real analytic functions is used; Sections 5 and 6 call on the implicit function theorem as well.

4.3 The induction

Proof of Theorem 1.2. Openness of $\mathcal{A}(f)$ is Lemma 4.1 and holds for every f . Density is proved by induction on the derivation of f in Definition 2.1.

Constants and the identity. Here $\mathcal{A}(f) = \mathbb{R}$.

Sums and products. If $\mathcal{A}(f)$ and $\mathcal{A}(g)$ are dense then so is $\mathcal{A}(f) \cap \mathcal{A}(g)$, because a finite intersection of dense open sets is dense. Analyticity is preserved by sums and products pointwise, so $\mathcal{A}(f) \cap \mathcal{A}(g) \subseteq \mathcal{A}(f + g)$ and likewise for fg ; a superset of a dense set is dense.

Negation. $\mathcal{A}(f) \subseteq \mathcal{A}(-f)$.

The unary clauses. Apply Lemma 4.2 with the pair (f, g) and the finite bad set B read off from Lemma 3.1: $B = \emptyset$ for $\exp$, $\sin$, $\cos$, $\arctan$; $B = \{0\}$ for $y \mapsto y^{-1}$, for $\log$, and for $y \mapsto y^r$; and $B = \{-1, 1\}$ for $\arcsin$. $\square$

Remark 4.4 (Sharpness). Density cannot be strengthened to “ $\mathcal{A}(f) = \mathbb{R}$ ”: the elementary functions $x \mapsto x^{-1}$, $x \mapsto |x|$ and $x \mapsto \arcsin x$ have analytic loci $\mathbb{R} \setminus \{0\}$, $\mathbb{R} \setminus \{0\}$ and $\mathbb{R} \setminus \{-1, 1\}$. Nor can “dense open” be strengthened to “open with finite complement”: the elementary function $x \mapsto (\sin x)^{-1}$ omits the infinite discrete set $\pi\mathbb{Z}$. What density does give for free is that the complement is closed and nowhere dense, being the complement of a dense open set. It need not be *discrete*, and one more

reciprocal settles that: $x \mapsto (\sin(1/x))^{-1}$ is elementary — it is `IsElementary.id.inv.sin.inv` — and its analytic locus omits every $1/(k\pi)$ with $k \in \mathbb{Z} \setminus \{0\}$, together with 0, where the function is not even continuous. Those points accumulate at 0. So “nowhere dense” is the correct description of the exceptional set and cannot be improved to “discrete”.

5 Algebraic functions over the elementary functions

Definition 2.1 reaches principal n -th roots on nonnegative inputs for positive n , and classical odd roots on all of $\mathbb{R}$, but not a general algebraic function. This section removes that restriction. It does so without enlarging $\mathcal{E}$, for a reason worth stating: an algebraic clause inside the inductive definition would cost Proposition 2.2. Composing a branch y with an elementary h produces a branch over the coefficients $a_i \circ h$, but it is a *continuous* branch only if h is continuous, and elementary functions need not be — $x \mapsto x^{-1}$ is not. Rather than weaken the closure theorem, we prove the conclusion directly.

Theorem 5.1 (Branches of polynomial equations). *Let $U \subseteq \mathbb{R}$ be open, let $a_0, \dots, a_n$ be real analytic on U with a_n nowhere zero on U , and let y be continuous on U with*

$$\sum_{i=0}^n a_i(x) y(x)^i = 0 \quad (x \in U).$$

Then y is real analytic at some point of every nonempty open subset of U .

Density is again the right conclusion, and for the same kind of reason as before: $y = |\cdot|$ is a continuous branch of $y^2 - x^2 = 0$ on all of $\mathbb{R}$, with $a_0 = -x^2$, $a_1 = 0$, $a_2 = 1$, and it is not analytic at the origin.

Proof architecture. The implicit function theorem applies only where $\partial P/\partial y \neq 0$, and the classical way to control the locus where it vanishes is the discriminant. `MathLib` has no usable discriminant theory — `Polynomial1.discr` carries no bridge to separability, and there is no continuity of roots in the coefficients — so we induct on the degree instead, which needs neither.

Write g for $\partial P/\partial y$ evaluated along the branch and split U into $V = \{x \in U : g(x) \neq 0\}$, which is open, and $W = U \setminus \overline{V}$, which is also open. On V the implicit function theorem applies directly. On W we have $g \equiv 0$, so y satisfies the *derivative* equation, of degree $n - 1$, whose leading coefficient $n a_n$ is still nowhere zero; the inductive hypothesis applies to it. Finally V together with a dense subset of W is dense in U , because an open set disjoint from V is disjoint from $\overline{V}$ and hence contained in W . The set W absorbs the *interior* of the degenerate locus $\{g = 0\}$; what the argument gives up is the boundary $\partial V \cap U$, and that is exactly why the conclusion is density rather than analyticity everywhere. For $y = |\cdot|$ the degenerate locus is $\{0\}$, so $V = \mathbb{R} \setminus \{0\}$, $\overline{V} = \mathbb{R}$, $W = \emptyset$, and the origin is conceded to the splitting step — correctly, since $|\cdot|$ is not analytic there.

The induction terminates because the degree drops. Its base case $n = 0$ is vacuous rather than positive: the equation reads $a_0(x) = 0$ while the leading coefficient hypothesis says $a_0(x) \neq 0$, so the region is empty.

The implicit function step is `MathLib's ContDiffAt.implicitFunction`, which is stated for every smoothness exponent in $\mathbb{N}_0 \cup \{\infty, \omega\}$; instantiating at ω and converting with `ContDiffAt.analyticAt` gives an analytic implicit function theorem. Its one nontrivial hypothesis, invertibility of the partial derivative as a continuous linear map, is discharged in one variable by `HasDerivAt.hasFDerivAt_equiv` together with uniqueness of the Fréchet derivative. $\square$

Corollary 5.2. *Let $U \subseteq [0, 1]$ be nonempty and open, let $a_0, \dots, a_n$ be elementary, let a_n vanish nowhere on U , and let y be continuous on U with $\sum_{i \leq n} a_i(x) y(x)^i = 0$ for every $x \in U$. Then y does not agree with the Fabius function on U ; in particular not on $(0, 1)$.*

Proof. Each a_i has dense open analytic locus by [Theorem 1.2](#). The intersection W of the $n + 1$ of them is a finite intersection of dense open sets, hence dense and open, so $U \cap W$ is nonempty and open, and [Theorem 5.1](#) applies to it: y is analytic at some $x \in U \cap W \subseteq U$. Now argue as in [Corollary 1.3](#). $\square$

Every hypothesis on the branch in [Corollary 5.2](#) is confined to U , and that is what makes the corollary say what one wants it to say. A version with global hypotheses would be much weaker than it looks. For the standard Bring equation $y^5 - y - x = 0$, a global continuous branch would be a continuous right inverse of $h(w) = w^5 - w$. Such a right inverse is injective; its image is an interval, and $h(y(x)) = x$ forces that image to be unbounded in both directions, hence all of $\mathbb{R}$. It would therefore force h itself to be injective, a contradiction. The localized statement nevertheless reaches the positive real branch over $(0, 1)$, an algebraic function not expressible by radicals.

The corollary is genuinely stronger than [Corollary 1.3](#) in that direction, and independent of it in the other: it says nothing about $\exp$ or $\log$ of an algebraic function, because it is a single algebraic step over elementary coefficients rather than a closure. A tower that interleaves algebraic extensions with exponentials and logarithms at several levels is covered only when it can be presented in one of the two forms.

6 The inverse, and why inverting cannot help

F is a homeomorphism of $[0, 1]$ onto itself, so it has a continuous inverse. We write $F^{-1}: \mathbb{R} \rightarrow \mathbb{R}$ for the formal development's clamped totalization: the inverse on $[0, 1]$, the constant 0 on $(-\infty, 0]$, and the constant 1 on $[1, \infty)$. Everything above concerned F ; this section treats that totalization, and then closes the class under the admissible inverse-branch rule.

In the formalization this function is `Fabius.fabiusInv`, defined by `Set.IccExtend` of the order isomorphism. The extension is continuous, so nothing is hidden in the gluing; its constant tails are what the endpoint half of [Theorem 6.2](#) exploits.

6.1 A right inverse is an implicit branch

The observation that organizes this section is that no new analysis is needed. The equation

$$h(g(x)) = x$$

says exactly that g is a continuous branch of $P(x, z) = h(z) - x = 0$, an implicit equation whose partial derivative in z is h' . So the analytic implicit function theorem quoted in [Section 5](#) — already proved there, for an unrelated purpose — delivers the analytic inverse function theorem with nothing further to check.

Theorem 6.1 (Analytic inverse function theorem). *If h is analytic at $g(x_0)$ with $h'(g(x_0)) \neq 0$, g is continuous at x_0 , and $h \circ g$ is the identity near x_0 , then g is analytic at x_0 .*

6.2 The inverse of the Fabius function

Theorem 6.2. *F^{-1} is real analytic at no point of $[0, 1]$; its analytic locus is exactly $\mathbb{R} \setminus [0, 1]$.*

Proof architecture. In the interior, run [Theorem 6.1](#) backwards. If F^{-1} is analytic at $y_0 \in (0, 1)$ then F , being its inverse near that point, is analytic at $F^{-1}(y_0) \in (0, 1)$ — provided $(F^{-1})'(y_0) \neq 0$. It has no critical point anywhere in the interior. This is the one place where F' enters at all — everywhere else F is used through its continuity and through its failure to be analytic — and what is needed of it

is only that $F' > 0$ on $(0, 1)$. Given that, the easy half of the inverse function theorem — the half that assumes the inverse already exists — yields

$$(F^{-1})'(y) = \frac{1}{F'(F^{-1}(y))} \quad (0 < y < 1),$$

a nonzero real number. The core inverse module formalizes this reciprocal derivative, its strict positivity, and the stronger fact that F^{-1} is C^∞ throughout $(0, 1)$. [Theorem 1.1](#) then supplies the contradiction.

The direction deserves to be stated, because the shorter route is circular. Differentiating $F^{-1} \circ F = \text{id}$ to obtain $(F^{-1})'(F(x)) \cdot F'(x) = 1$ presupposes that F^{-1} is differentiable, which is exactly what is at issue; strict positivity of F' is what supplies it.

At the endpoints the argument is different, and simpler. F^{-1} is identically 0 to the left of 0; an analytic germ agreeing with 0 on a set with an accumulation point vanishes identically, so F^{-1} would vanish just to the right of 0 as well, contradicting strict monotonicity. Symmetrically at 1. $\square$

Corollary 6.3. *No elementary function agrees with F^{-1} on any subset of $[0, 1]$ with nonempty interior; in particular the restriction of F^{-1} to $(0, 1)$ is not elementary.*

One might have hoped that inverting improves matters. F is flat to infinite order at the endpoints, so F^{-1} is steep there, and steepness is not obviously an obstruction to being elementary — the signed real cube root $x \mapsto (x/|x|)|x|^{1/3}$ is steep at the origin and perfectly elementary. [Theorem 6.2](#) says the hope is misplaced, and the reason is structural rather than quantitative: analytic functions are closed under inversion wherever the derivative survives, so an analytic inverse would hand analyticity straight back.

6.3 Closing the class under inversion

Nothing about the argument was special to F . The same reasoning shows that the invariant of [Theorem 1.2](#) survives the admissible inverse-branch rule below.

Theorem 6.4. *Let f be analytic on a dense set and let g be continuous on a nonempty open U with $f(g(y)) = y$ there. Then g is analytic at some point of U .*

Proof architecture. Two facts. A continuous right inverse is injective, and a continuous injection carries an interval to a set containing an interval — the intermediate value theorem — so the image $g(U)$ is *fat* and the dense set $\mathcal{A}(f)$ must meet it. And on that image f is injective, being inverse to g , so f cannot be locally constant there; its derivative therefore fails to vanish somewhere in $\mathcal{A}(f) \cap g(U)$. At such a point [Theorem 6.1](#) applies, and transporting back through the branch gives a point of U at which g is analytic. $\square$

Applied to every nonempty open subset at once, this says that a continuous inverse branch is analytic on a relatively dense subset of its domain, and the domain need not be open: `analyticDenseOn_of_rightInverse`. For a domain with empty interior the statement is vacuous, which is the correct reading — there is no open set on which to test.

The conclusion is “analytic somewhere in U ”, not “analytic on U ”, and that is not slack in the argument: the real cube root is a continuous right inverse of the entire function $x \mapsto x^3$ on all of $\mathbb{R}$ and is not analytic at the origin. In the formalization the right inverse has to be written $t \mapsto (t/|t|)|t|^{1/3}$, since `Real.rpow` makes $(-8)^{1/3} = 1$ ([Warning 2.3](#)); that this signed form really does cube back to t is `Fabius.signedRoot_pow`. Critical points of f do produce singularities of the inverse. It is the same phenomenon that made density, rather than analyticity everywhere, the right invariant in [Theorem 1.2](#).

Definition 6.5. $\mathcal{E}^{-1}$ is the smallest class containing $\mathcal{E}$, closed under the operations of Definition 2.1, and closed under the following rule: if $f \in \mathcal{E}^{-1}$, U is open, g is continuous on U with $f(g(y)) = y$ for $y \in U$, and g is analytic on the interior of the complement of U , then $g \in \mathcal{E}^{-1}$.

Two features of the rule are deliberate, and the Lambert W function shows why each is needed. It is local in U : W satisfies its defining identity $W(y)e^{W(y)} = y$ only on $[-1/e, \infty)$, so a rule demanding a global identity would not reach it. And it asks for analyticity on the *interior* of the complement rather than at every point outside U : the branch point $-1/e$ sits on the boundary of the natural $U = (-1/e, \infty)$, and W is certainly not analytic there. Dropping that boundary costs nothing, since the boundary of an open set is nowhere dense, and for W the constant extension then satisfies the analyticity clause on all of $(-\infty, -1/e)$.

The Lean result is a conditional criterion for any supplied total function satisfying these hypotheses. Since `Mathlib` does not define W , the formalization neither constructs a standard real branch nor verifies a particular totalization of it.

That clause is a condition on the chosen extension of the branch past U , not a fact that comes for free. Extension by a constant always meets it, and that is the honest reading of the rule: a branch is admitted together with a declaration of how it is continued. Other conventions also meet it without being constant — `Mathlib` sets $\log y = \log |y|$, which is analytic off the origin. At the other extreme $U = \emptyset$ reduces the clause to “ g is entire”, so the rule also adjoins every everywhere-analytic function; harmlessly, since those already have dense analytic locus, which is all the invariant asks. Inversion may be applied at any depth: to members of $\mathcal{E}$, to branches of those, and so on without limit.

One asymmetry with $\mathcal{E}$ deserves stating, since Section 2 made a point of it. Composition is a *theorem* for $\mathcal{E}$ (Proposition 2.2), proved by inducting on the derivation of the outer function, because every clause of Definition 2.1 commutes with precomposition. That induction does not replay for $\mathcal{E}^{-1}$: the inverse clause does not commute with precomposition, since g being a branch of $f(z) = x$ says nothing about $g \circ h$ being a branch of anything. So $\mathcal{E}^{-1}$ is not claimed to be closed under composition, and Theorem 6.6 does not need it — the induction there is over the derivation, and every clause of Definition 6.5 preserves dense analyticity on its own.

Theorem 6.6. *Every member of $\mathcal{E}^{-1}$ is analytic on a dense set.*

Proof. Induction. The elementary generators are Theorem 1.2; the operations are as in Section 4; and the inverse rule splits on whether the test set meets U . Given a nonempty open B : if $B \cap U \neq \emptyset$ then Theorem 6.4 applies to the nonempty open set $B \cap U$ and produces a point of B at which g is analytic; and if $B \cap U = \emptyset$ then $B \subseteq U^c$, so $B \subseteq \text{int}(U^c)$ because B is open, and every point of B will do by the rule’s analyticity clause. $\square$

Corollary 6.7. *Neither F nor F^{-1} agrees with any member of $\mathcal{E}^{-1}$ on any subset of $[0, 1]$ with nonempty interior. In particular neither is representable on $(0, 1)$ by elementary functions, elementary operations, and any finite tower of totalized inverse branches satisfying Definition 6.5.*

It is worth being explicit about what Corollary 6.7 does and does not rest on. The inverse need not be two-sided: a continuous right inverse on an open set is enough. On the other hand, the conditional Lambert- W constructor also requires the chosen totalization to be analytic on the interior of that open set’s complement. The branch-domain boundary is deliberately excluded from this side condition. Because `Mathlib` does not define W , the formalization neither constructs a standard real branch nor verifies any particular totalization.

7 The formal development

7.1 Correspondence with the Lean names

All statements below live in the namespace `Fabius`.

Object or statement	Lean name	Module
Elementary functions (Definition 2.1)	<code>IsElementary</code>	<code>ElementaryFunction</code>
Closure under composition (Proposition 2.2)	<code>IsElementary.comp</code>	<code>ElementaryFunction</code>
Derived closures: $-$, $/$, x^n , $\sqrt{\cdot}$, $ \cdot $, $\tan$, $\sinh$, $\cosh$, $\tanh$, $\arccos$, arsinh , polynomials	<code>IsElementary.sub</code> , <code>.div</code> , <code>.npow</code> , <code>.sqrt</code> , <code>.abs</code> , <code>.tan</code> , <code>.sinh</code> , <code>.cosh</code> , <code>.tanh</code> , <code>.arccos</code> , <code>.arsinh</code> , <code>.polynomial</code>	<code>ElementaryFunction</code>
Analytic locus $\mathcal{A}(f)$	<code>analyticLocus</code>	<code>ElementaryFunction</code>
Openness (Lemma 4.1)	<code>isOpen_analyticLocus</code>	<code>ElementaryFunction</code>
Generators (Lemma 3.1)	<code>analyticAt_log_of_ne_zero</code> , <code>analyticAt_rpow_const</code> , <code>analyticAt_arctan</code> , <code>analyticAt_arcsin</code> ; for $x \mapsto x^{-1}$, <code>exp</code> , <code>sin</code> , <code>cos</code> the Mathlib lemmas <code>analyticAt_inv</code> , <code>analyticAt_rexp</code> , <code>Real.analyticAt_sin</code> , <code>Real.analyticAt_cos</code> are used directly	<code>ElementaryFunction</code>
Auxiliary: analyticity of $\sqrt{\cdot}$ off 0 (not used in the induction, which routes $\sqrt{\cdot}$ through <code>rpow</code>)	<code>analyticAt_sqrt_of_ne_zero</code>	<code>ElementaryFunction</code>
Signed root, and its defining property (Warning 2.3)	<code>IsElementary.signedRpow</code> , <code>signedRoot_pow</code>	<code>ElementaryFunction</code>
Variable exponents at a positive base, and at any nonvanishing base	<code>IsElementary.rpow_of_pos</code> , <code>IsElementary.rpow_of_ne_zero</code> , <code>eqOn_rpow_of_pos</code>	<code>ElementaryFunction</code>
Membership of the named functions	<code>isElementary_exp</code> , <code>_log</code> , <code>_sin</code> , <code>_cos</code> , <code>_tan</code> , <code>_arcsin</code> , <code>_arccos</code> , <code>_arctan</code> , <code>_sinh</code> , <code>_cosh</code> , <code>_tanh</code> , <code>_arsinh</code> , <code>_sqrt</code> , <code>_abs</code> , <code>_inv</code> , <code>_rpow</code> , <code>_id</code>	<code>ElementaryFunction</code>
Key lemma (Lemma 4.2)	<code>dense_analyticLocus_comp</code>	<code>ElementaryFunction</code>
Density (Theorem 1.2)	<code>IsElementary.dense_analyticLocus</code>	<code>ElementaryFunction</code>
Analytic at a point of any nonempty open set	<code>IsElementary.exists_analyticAt_of_isOpen</code>	<code>ElementaryFunction</code>
Exceptional set closed with empty interior	<code>isClosed_compl_analyticLocus</code> , <code>IsElementary.interior_compl_analyticLocus</code>	<code>ElementaryFunction</code>
Analytic implicit function theorem in two real variables	<code>analyticAt_implicitFunction</code> , <code>analyticAt_of_continuous_branch</code>	<code>AlgebraicBranch</code>
Branches of polynomial equations (Theorem 5.1)	<code>analyticDenseOn_of_algebraic</code> , <code>dense_setOf_analyticAt_of_algebraic</code> , <code>branchPoly</code> , <code>AnalyticDenseOn</code> , <code>analyticDenseOn_of_inter_open_dense</code> , <code>AnalyticDenseOn.not_eqOn_of_forall_not_analyticAt</code>	<code>AlgebraicBranch</code>
Algebraic functions over $\mathcal{E}$ (Corollary 5.2)	<code>analyticDenseOn_of_isElementary_coeffs</code> , <code>exists_analyticAt_of_isElementary_coeffs</code> , <code>not_algebraicBranch_eqOn</code> , <code>canonical_fabius_not_algebraicBranch_on_Ioo</code>	<code>NotElementary</code>
The obstruction, stated once (proof of Corollary 1.3)	<code>not_eqOn_of_dense_analyticLocus</code>	<code>ElementaryFunction</code>
Analytic inverse function theorem (Theorem 6.1)	<code>analyticAt_of_rightInverse</code>	<code>InverseBranch</code>

Object or statement	Lean name	Module
Inverse branches are analytic somewhere (Theorem 6.4)	<code>exists_analyticAt_of_rightInverse,</code> <code>analyticDenseOn_of_rightInverse</code>	<code>InverseBranch</code>
The class $\mathcal{E}^{-1}$ (Definition 6.5 , Theorem 6.6)	<code>IsElementaryOrInverse, IsElementa</code> <code>ryOrInverse.dense_analyticLocus</code>	<code>InverseBranch</code>
Conditional Lambert- W inverse-branch criterion	<code>isElementaryOrInverse_of_lambertW</code>	<code>InverseBranch</code>
Interior calculus and smoothness of F^{-1}	<code>fabiusInv_mem_Ioo,</code> <code>fabiusInv_hasDerivAt,</code> <code>hasDerivAt_fabiusInv,</code> <code>deriv_fabiusInv, deriv_fabiusInv_</code> <code>eq_inv_two_mul_rvachevUp,</code> <code>deriv_fabiusInv_pos,</code> <code>fabiusInv_contDiffOn_Ioo</code>	<code>FabiusInverse</code>
Exact global differentiability and smoothness locus of the totalized F^{-1}	<code>fabiusInv_contDiffAt, fabiusInv_no</code> <code>t_differentiableAt_zero, fabiusIn</code> <code>v_not_differentiableAt_one,</code> <code>fabiusInv_differentiableAt_iff,</code> <code>fabiusInv_contDiffAt_iff,</code> <code>fabiusInv_contDiffAt_infty_iff,</code> <code>fabiusInv_contDiffOn_compl_endpo</code> <code>ints</code>	<code>FabiusInverse</code>
F^{-1} has no critical point, and is nowhere analytic (Theorem 6.2)	<code>deriv_fabiusInv_ne_zero,</code> <code>fabiusInv_not_analyticAt,</code> <code>fabiusInv_analyticAt_iff</code>	<code>InverseNotElementary</code>
F^{-1} is not elementary (Corollary 6.3)	<code>canonical_fabiusInv_not_isElemen</code> <code>tary_on_Ioo,</code> <code>fabiusInv_not_isElementary, not_eq</code> <code>On_fabiusInv_of_analyticDenseOn,</code> <code>not_eqOn_fabiusInv_of_dense_anal</code> <code>yticLocus</code>	<code>InverseNotElementary</code>
Neither is reachable by the inverse-branch closure (Corollary 6.7)	<code>not_isElementaryOrInverse_eqOn_f</code> <code>abius, not_isElementaryOrInverse_</code> <code>eqOn_fabiusInv (general U),</code> <code>canonical_fabius_not_isElementar</code> <code>yOrInverse_on_Ioo,</code> <code>canonical_fabiusInv_not_isElemen</code> <code>taryOrInverse_on_Ioo ($U = (0, 1)$)</code>	<code>InverseNotElementary</code>
Nowhere analyticity (Theorem 1.1)	<code>canonical_fabius_not_analyticAt,</code> <code>canonical_fabius_analyticAt_iff</code>	<code>NowhereAnalytic</code>
Main result (Corollary 1.3)	<code>canonical_fabius_not_isElementar</code> <code>y_eqOn_of_interior_nonempty (general</code> <code>U), canonical_fabius_not_isElemen</code> <code>tary_on_Ioo ($U = (0, 1)$)</code>	<code>NotElementary</code>
Interior form of the obstruction	<code>IsElementary.not_eqOn_of_interio</code> <code>r_nonempty; not_isElementary_eqOn_</code> <code>of_interior_nonempty, not_algebrai</code> <code>cBranch_eqOn_of_interior_nonempty</code>	<code>ElementaryFunction;</code> <code>NotElementary</code>
F is not elementary	<code>canonical_fabius_not_isElementary</code>	<code>NotElementary</code>
General open subset of $[0, 1]$	<code>not_isElementary_eqOn</code>	<code>NotElementary</code>
Rvachev's up and the signed extension	<code>rvachev_not_isElementary,</code> <code>extendedFabius_not_isElementary</code>	<code>NotElementary</code>

All modules are under `Analysis/FabiusFunction/Lean/FabiusFunction/`.

7.2 Ingredients taken from Mathlib

The formalization is short because `Mathlib` supplies the analytic infrastructure that the argument consumes.

- `AnalyticAt.eventually_eq_zero_or_eventually_ne_zero` and its two-function form:

the isolated-zeros theorem for a function of one variable over a nontrivially normed field, which is what powers Lemma 4.2. It applies verbatim to $f: \mathbb{R} \rightarrow \mathbb{R}$.

- `isOpen_analyticAt`: the analytic locus is open, which is Lemma 4.1 verbatim. (The pointwise form, analyticity propagating from a point to a neighbourhood, is `AnalyticAt.eventually_analyticAt`; the development uses the set-level statement.)
- `ContDiffAt.analyticAt` together with the family of `contDiffAt_` . . . lemmas stated for exponents in $\mathbb{N}_0 \cup \{\infty, \omega\}$: this is how Lemma 3.1 is obtained, including the two facts that are otherwise awkward over $\mathbb{R}$, namely analyticity of $\arctan$ everywhere and of $\arcsin$ off $\{\pm 1\}$.
- `ContDiffAt.implicitFunction`, instantiated at the analytic exponent ω , supplies the local analytic implicit-function theorem used in Theorem 5.1.

7.3 Statement of the main theorem in Lean

```
theorem canonical_fabius_not_isElementary_on_Ioo :
  not (exists g : ℝ -> ℝ, IsElementary g and
    Set.EqOn g (fabiusReal fabius) (Set.Ioo 0 1))
```

(rendered here in ASCII; the source uses the usual notation). The development contains no `sorry`, no axiom and no `opaque`, and its axiom set is the Mathlib default `propext`, `Classical.choice`, `Quot.sound`.

8 Discussion

8.1 Why smoothness cannot decide the question

The Fabius function is C^∞ on $\mathbb{R}$, and its derivatives satisfy the exact self-similar relation $F'(x) = 2F(2x)$ on $[0, 1/2]$. Any proof of non-elementarity must therefore distinguish F from the elementary functions by a property strictly finer than differentiability of every order. Real analyticity is the natural candidate, and Theorem 1.2 says it is a sufficient one. It is also, in a precise sense, the *only* such candidate we know how to extract from Definition 2.1 directly. Nothing in this document rules out a different separating property; what it does show is that analyticity suffices, and that it is available at the modest cost of Lemma 4.2.

The same is true of the inverse, and for the same reason. Bootstrapping the reciprocal formula of Theorem 6.2 through its own derivative shows F^{-1} to be C^∞ on $(0, 1)$ — this is `fabiusInv_contDiffOn_Ioo` — so F^{-1} is a second function that is smooth to every order on an interval and real analytic at no point of it. Smoothness separates neither of them from $\mathcal{E}$; analyticity separates both.

8.2 The role of the endpoints

Theorem 1.1 includes the endpoints 0 and 1, where F is flat to infinite order. It is worth recording that Corollary 1.3 does *not* need them. A set that is open in $\mathbb{R}$ and contained in $[0, 1]$ is automatically contained in $(0, 1)$ — an $\mathbb{R}$ -open set containing 0 contains negative numbers — so the witness produced by Theorem 1.2 is always an interior point, and the endpoint case of Theorem 1.1 is never mathematically reached. (The proof term still cites it: the applications call `fabius_not_analyticAt`, which is the combined statement over all of $[0, 1]$ and case-splits on the endpoint internally.) The same is true if one reads “open” in the relative topology of $[0, 1]$: a nonempty relatively open subset still contains a nonempty $\mathbb{R}$ -open subinterval of $(0, 1)$.

What the endpoints do buy is exactness. $\mathcal{A}(F)$ is precisely $\mathbb{R} \setminus [0, 1]$: outside the unit interval F is locally constant, hence analytic, so the nowhere-analytic locus is as large as it can be and no larger.

8.3 What a stronger theorem would require

Two natural strengthenings are visible from here.

1. *Interleaved towers.* Replacing $\mathcal{E}$ by the class of functions lying in an *elementary* differential-field extension of the rational functions in Liouville's sense — allowing arbitrary algebraic branches alongside exponentials and logarithms, at any depth — would subsume both Definition 2.1 and Section 5. (The wider *Liouvillian* class also admits antiderivatives, and so contains erf, Ei and Si, which are not elementary; that class is not what is meant here.) Each of the two ingredients is now formalized separately; what is missing is the closure that lets them alternate. The obstruction is the one identified at the head of Section 5: composing an algebraic branch with an elementary function needs the inner function to be continuous. Repairing that needs the fact that a nonconstant real analytic function on an interval has image containing an interval, so that the preimage of a dense open set is again dense — true, classical, and not formalized here.
2. *Differential algebraicity.* A stronger and genuinely different statement is that F satisfies no algebraic differential equation $P(x, F, F', \dots, F^{(n)}) = 0$ with polynomial P . Such a statement does not follow from nowhere analyticity — differentially algebraic functions need not be analytic — and is not addressed here.